

UNIT THREE UNCONSTRAINED OPTIMIZATION

Section 3.1 Functions of One Independent Variable

Some objective functions involve constraints and others do not. Functions which do not involve constraints are referred to as unconstrained functions and the process of optimization is said to be unconstrained or free optimization.

Given the function $y = f(x)$ which is continuous and differentiable, it is said to have a maximum value at a point where it changes from an increasing to decreasing function where as it is said to have a minimum value at the point where it changes from decreasing to increasing functions. The values of x at which the function is at its minimum or maximum point are known as critical values. The given function should satisfy two conditions in order to decide about maximum and minimum value at a particular point. These conditions are called order conditions.

a) Conditions for minimum value

1. First order condition

$$f'(x) = 0$$

(Necessary condition)

2. Second order condition

$$f''(x) > 0$$

(Sufficient condition)

b) Conditions for Maximum value

1. First order condition

$$f'(x) = 0$$

(Necessary condition)

2. Second order condition

$$f''(x) < 0$$

(Sufficient Condition)

However, the second derivative of the function may be equal to zero in some cases. When $f''(x) = 0$, the second derivative test would be inconclusive. Thus, we should take the successive derivative test to determine whether the function is at its extremum point or the point of inflection. Evaluated at a critical point if the first non - zero value of a higher order derivative is an odd numbered derivative, then the function is at the point of inflection. If the first non - zero value of a higher order derivative is an even numbered derivative, the function is at it's relative extremum with positive value the derivative shows the relative minimum and with negative value the derivative indicates the relative maximum point.

Example

Find the minimum and maximum values of the function $f(x) = 3x^4 - 10x^3 + 6x^2 + 5$

Solution

The first order condition is $f'(x) = 0$. We can determine the critical values using this condition.

Thus, $f'(x) = 12x^3 - 30x^2 + 12x = 0$

$$3x(4x^2 - 10x + 4) = 0$$

$$3x[2x(2x - 1) - 4(2x - 1)] = 0$$

$$3x[(2x - 4)(2x - 1)] = 0$$

$$3x = 0 \quad \text{or} \quad 2x - 4 = 0 \quad \text{or} \quad 2x - 1 = 0$$

Thus the critical values are $x = 0$ or $x = 2$ or $x = \frac{1}{2}$.

We should test the second order condition at these points to know whether the function is at its relative maximum or minimum point.

$$f''(x) = 36x^2 - 60x + 12$$

At $x = 0$, $f''(x) = 36(0)^2 - 60(0) + 12 = 12 > 0$. Thus, the function is at its relative minimum point at $x = 0$.

$$\begin{aligned} \text{At } x = 2, f''(x) &= 36(2)^2 - 60(2) + 12 \\ &= 144 - 120 + 12 \end{aligned}$$

$f''(x) = 36 > 0$. Thus, the function is at its relative minimum point at $x = 2$.

$$\begin{aligned} \text{At } x = \frac{1}{2}, f''(x) &= 36\left(\frac{1}{2}\right)^2 - 60\left(\frac{1}{2}\right) + 12 \\ &= 36\left(\frac{1}{4}\right) - 30 + 12 \\ &= 9 - 30 + 12 = -9 < 0 \end{aligned}$$

Therefore, the function is at its relative maximum point when $x = \frac{1}{2}$.

3.1.1 Economic Applications

Revenue functions

As you remember revenue represents the amount of money that the firm generates either from the sale of the products or providing services. Therefore,

$$\text{Total Revenue} = \text{Price} \times \text{Quantity Sold}$$

It can be written as

$$TR = P \times Q$$

Where P is price and Q is quantity.

The firm can maximize its total revenue when

$$\frac{dTR}{dQ} = 0 \quad \text{or} \quad MR = 0 \quad (\text{First order condition})$$

$$\text{and} \quad \frac{d^2TR}{dQ^2} < 0 \quad \text{or} \quad \frac{d(MR)}{dQ} < 0 \quad (\text{Second order condition})$$

Example

1. A farmer wants to send his perishable product to a city market as soon as possible. He has estimated that he can send now 1.5 tones of the product per day and can get a price of 2,500 birr per tones. If he waits, he can get a price increase of 20 Birr per tones per day but the quantity of the product to be supplied will be reduced by 0.01 tones per day. For how many days should he wait so that his revenue becomes a maximum?

Solution

Let x be the number of days waiting

$$\begin{aligned} \text{Total revenue (TR)} &= (2,500 + 20x)(1.5 - 0.01x) \\ &= 3,750 - 25x + 30x - 0.2x^2 \end{aligned}$$

$$TR = 3,750 + 5x - 0.2x^2$$

Revenue is maximized when

$$\frac{dTR}{dx} = 0 \quad \text{and} \quad \frac{d^2TR}{dx^2} < 0$$

$$\begin{aligned} \text{Thus, } \frac{dTR}{dx} &= 5 - 0.4x = 0 \\ -0.4x &= -5 \\ x &= 12.5 \end{aligned}$$

$$\text{At } x = 12.5, \frac{d^2TR}{dx^2} = -0.4 < 0$$

Therefore, the farmer should wait for 12.5 days to maximize he's revenue.

2. Suppose a 200- room hotel in Addis Ababa will rent all its rooms when it charges 125 Birr per night per room. However from past experience, the manager knows that for each 5 birr increase in rate per night per room, 4 rooms remain unoccupied per night. What rent per room will maximize total revenue per night?

Solution

Let x be increase in nightly rent per room.

$$\begin{aligned} TR &= (200 - 4x)(125 + 5x) \\ &= 25,000 + 1,000x - 500x - 20x^2 \\ TR &= 25,000 + 500x - 20x^2 \end{aligned}$$

$$\text{Total revenue is maximized when } \frac{dTR}{dx} = MR = 0 \text{ and } \frac{d^2TR}{dx^2} < 0$$

$$\begin{aligned} MR &= 500 - 40x = 0 \\ -40x &= -500 \\ x &= 12.5 \end{aligned}$$

$$\text{When } x = 12.5, \frac{d^2TR}{dx^2} = -40 < 0$$

Thus, total revenue is maximized when $x = 12.5$

The rent per room per night that should be charged to which maximizes revenue is

$$\begin{aligned} \text{Rent} &= 125 + 5(12.5) = 125 + 6.5 \\ &= 187.5 \text{ Birr per room per night} \end{aligned}$$

Profit functions

As we know that the main objective of every firm is profit maximization; therefore, it wants to know the level of output which maximizes profit.

$$\text{Total profit} = \text{Total revenue} - \text{Total cost}$$

$$\Pi = TR - TC$$

Where Π represents total profit, TR is total revenue and TC is total cost.

We have discussed above that there are first orders and second order conditions for maximizing profit. The first order condition that must be fulfilled for maximizing profit is that the slope of total revenue (marginal revenue) has to be equal to the slope of total cost (marginal cost). That is

$$\frac{d\Pi}{dQ} = 0 \Rightarrow \frac{dTR}{dQ} - \frac{dTC}{dQ} = 0$$

$$MR - MC = 0$$

$$MR = MC$$

The second condition that has to be satisfied is that the slope of marginal revenue must be less than the slope of marginal cost. In other words, the marginal cost curve has to cross the marginal revenue curve from below. That is

$$\frac{d^2\Pi}{dQ^2} < 0 \Rightarrow \frac{d^2TR}{dQ^2} - \frac{d^2TC}{dQ^2} < 0$$

$$\frac{d(MR)}{dQ} < \frac{d(MC)}{dQ}$$

$$\text{Slope of MR} < \text{Slope of MC}$$

Example

1. Suppose a monopolist has a demand curve $Q = 106 - 2P$ and average cost curve $AC = 5 + \frac{Q}{50}$, where P is price per unit and Q is the number of units of output. Determine the profit maximizing level of output and price of this monopolist.

Solution

As we know, total profit (Π) = Total Revenue (TR) - Total Cost (TC) but $TR = PQ$. Thus we should rewrite the demand function in the form of price expressed in terms of quantity. That is

$$2P = 106 - Q$$

$$P = 53 - \frac{1}{2}Q$$

$$\text{Thus, } TR = (53 - \frac{1}{2}Q)Q$$

$$TR = 53Q - \frac{1}{2}Q^2 \Rightarrow MR = 53 - Q$$

And

$$TC = AC(Q)$$

$$= (5 + \frac{Q}{50})Q$$

$$TC = 5Q + \frac{1}{50}Q^2 \Rightarrow MC = 5 + \frac{1}{25}Q$$

profit is maximized when

$$MR = MC$$

$$53 - Q = 5 + \frac{1}{25}Q$$

$$53 - 5 = Q + \frac{1}{25} Q$$

$$48 = \frac{26}{25} Q$$

$$Q = \frac{48(25)}{26}$$

$$Q = 46.15$$

However, this information is not sufficient enough to conclude that it is the profit maximizing level of output. Thus it must fulfill the following condition at this point.

$$\begin{aligned} \text{When } Q = 46.15, \frac{d^2 TR}{dQ^2} - \frac{d^2 TC}{dQ^2} &= -1 - \frac{1}{25} \\ &= \frac{-26}{25} < 0 \end{aligned}$$

Now we are confident enough to conclude that $Q = 46.15$ is the profit maximizing level of output of the monopolist as this level of output satisfies both of the above conditions.

The profit maximizing level of price is

$$\begin{aligned} P &= 53 - \frac{1}{2} (46.15) \\ &= 53 - 23.075 \\ P &= 29.93 \end{aligned}$$

Cost Functions

Do you remember the analysis of cost in your Microeconomics I study? Please try to re-read that material so that you will have a good understanding for this of this topic. As you remember costs represent the amount of expenditures that firm's incurred in the production process. It includes both implicit and explicit cost. Thus, firms want to minimize these costs.

Given Total Cost (TC) = $f(Q)$, where Q is output, Firms can minimize total cost if and only if

$$\begin{aligned} \text{i) } \frac{dTC}{dQ} &= 0 \Rightarrow MC = 0 \\ \text{ii) } \frac{d^2 TC}{dQ^2} &> 0 \Rightarrow \frac{d(MC)}{dQ} > 0 \end{aligned}$$

Example

1. Suppose the total cost of producing Q units of a certain product is described by the function $TC = 100,000 + 1,500Q + 0.4Q^2$ where TC is the total cost stated in Birr. Determine the amount of output which minimizes average cost.

Solution

$$\text{Average cost (AC)} = \frac{TC}{Q}$$

$$AC = \frac{100,000}{Q} + 1,500 + 0.4Q$$

AC is minimized when

$$\frac{dAC}{dQ} = 0 \quad \text{And} \quad \frac{d^2(AC)}{dQ^2} > 0$$

$$\begin{aligned}\frac{dAC}{dQ} &= \frac{-100,000}{Q^2} + 0.4 = 0 \\ -\frac{100,000}{Q^2} &= -0.4 \\ \frac{100,000}{0.4} &= Q^2 \\ Q^2 &= 250,000 \\ Q &= \pm 500 \text{ but output should be positive.}\end{aligned}$$

Therefore, $Q = 500$. But we should check the second order condition at this point to reach to our conclusion.

$$\begin{aligned}\text{When } Q = 500, \frac{d^2 AC}{dQ^2} &= \frac{-2(-100,000)}{Q^3} \\ &= \frac{200,000}{Q^3} = \frac{200,000}{(500)^3} > 0\end{aligned}$$

Thus, the level of output which minimizes average cost is $Q = 500$ units.

- Find the minimum point of the average cost function

$$Ac = 25 Q^{-1} + 0.1Q^2$$

Solution

Applying the same conditions

$$\begin{aligned}\frac{dAC}{dQ} &= -25 Q^{-2} + 0.2Q = 0 \\ -25Q^{-2} + 0.2Q &= 0 \\ -25Q^{-2} &= -0.2Q \\ \frac{25}{Q^2} &= 0.2Q \\ Q^3 &= 125\end{aligned}$$

Therefore, Ac is at its stationary point when $Q = 5$. The rate of change of the slope of AC with

respect to output is $\frac{d^2 AC}{dQ^2} = 50 Q^{-3} + 0.2$

$$\begin{aligned}\text{When } Q = 5, \frac{d^2 AC}{dQ^2} &= \frac{50}{5^3} + 0.2 \\ &= \frac{50}{125} + 0.2 \\ &= 0.4 + 0.2 = 0.6 > 0\end{aligned}$$

Thus, the second order condition for a minimum value of AC is satisfied when $Q = 5$.

The actual value of AC at its minimum point will be

$$25 Q^{-1} + 0.1Q^2 = \frac{25}{5} + 0.1(25) = 5 + 2.5 = 7.5$$

Section 3.2 Functions of Several Independent Variables

Until now we have discussed about the problem of optimization of an objective function with only one choice variable. Now let us turn our attention to develop away of finding the extreme values of an objective function which includes two or more choice variables. Then we will have the ability to solve problems such as determining the profit maximizing level of outputs for several commodities and the optimal combinations of several different inputs.

Given the function $z = f(x, y)$, the objective function z to be maximum or minimum, it must satisfy both of the order conditions.

The first order conditions are

$$\frac{\partial z}{\partial x} = 0 \quad \text{and} \quad \frac{\partial z}{\partial y} = 0$$

This means, the first order total differential of the function is zero ($dz = f_x dx + f_y dy = 0$).

However, there are two sets of second order conditions

$$\text{a) To be maximum } \frac{\partial^2 z}{\partial x^2} < 0, \text{ and } \frac{\partial^2 z}{\partial y^2} < 0 \text{ ----- (1)}$$

$$\text{b) To be minimum } \frac{\partial^2 z}{\partial x^2} > 0, \text{ and } \frac{\partial^2 z}{\partial y^2} > 0$$

The other second order condition for both to be at maximum and minimum value is

$$\left(\frac{\partial^2 z}{\partial x^2}\right) \left(\frac{\partial^2 z}{\partial y^2}\right) > \left(\frac{\partial^2 z}{\partial x \partial y}\right)^2$$

Alternatively we can determine the second order sufficient condition using the concept of total differential of the differential of the function which is denoted by

$$\begin{aligned} d^2 Z &= d(dZ) = \frac{\partial(f_x dx + f_y dy)}{\partial x} dx + \frac{\partial(f_x dx + f_y dy)}{\partial y} dy \\ d^2 Z &= (f_{xx} dx + f_{xy} dy) dx + (f_{yx} dx + f_{yy} dy) dy \text{ ----- (2)} \\ d^2 Z &= f_{xx} dx^2 + f_{xy} dy dx + f_{yx} dx dy + f_{yy} dy^2 \end{aligned}$$

But we know that $f_{xy} = f_{yx}$ (Young's Theorem)

$$d^2 Z = f_{xx} dx^2 + 2f_{xy} dx dy + f_{yy} dy^2 \text{ ----- (3)}$$

Then the second order condition for any value of dx and dy not both zero

$d^2 Z < 0$ indicates that the function is at its maximum whereas $d^2 Z > 0$ shows that the function at it minimum point. However, for any value of dx and dy ,

$$d^2 Z < 0 \text{ if and only if } f_{xx} < 0, f_{yy} < 0 \text{ and } f_{xx} f_{yy} - (f_{xy})^2 > 0.$$

$$d^2 Z > 0 \text{ if and only if } f_{xx} > 0, f_{yy} > 0 \text{ and } f_{xx} f_{yy} - (f_{xy})^2 > 0.$$

Given the quadratic function $U = ax_1^2 + bx_1x_2 + cx_2^2$, the discriminant, which is symmetric determinant, is formed by putting the coefficient of the squared variable on the principal diagonal and dividing the coefficient of the non squared term equally between the off-diagonal positions as follows

$$|D| = \begin{vmatrix} a & \frac{b}{2} \\ \frac{b}{2} & c \end{vmatrix}$$

Where $|D_1| = a$ the first principal minor of the discriminant and $|D_2| = ac - \frac{b^2}{4}$ is the second principal minor. The sign definiteness of the given function can be described in terms of sign restriction on these principal minors.

U is positive definite iff $|D_1| > 0$ and $|D_2| > 0$.

U is negative definite $|D_1| < 0$ and $|D_2| > 0$

For n-variable quadratic form of the function, the discriminant becomes

$$|D| = \begin{vmatrix} d_{11} & d_{12} & d_{13} & \dots & d_{1n} \\ d_{21} & d_{22} & d_{23} & \dots & d_{2n} \\ d_{31} & d_{32} & d_{33} & \dots & d_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & d_{n3} & \dots & d_{nn} \end{vmatrix}$$

It has n- number of principal minors. The necessary and sufficient condition for sign definiteness is that all principal minors must be greater than zero for positive definiteness and they have to alternate in sign as $|D_1|$ is negative for negative definiteness.

The total differential expressed in equation (3) above is in a quadratic form. As a result, the discriminant is a determinant that contains the second order partial derivatives as its elements. This determinant is referred to as Hessian determinant.

$$|H| = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$$

$|H_1| = f_{xx}$ And $|H_2| = f_{xx}f_{yy} - f_{yx}f_{xy}$ are the first and second principal minors respectively.

By now we can express the sign definiteness of d^2Z using the sign of these principal minors and thereby we are able to identify the second order sufficient conditions for the extremum of the function $Z = f(x, y)$ as

d^2Z is positive definite iff $|H_1| = f_{xx} > 0$ And $|H_2| = f_{xx}f_{yy} - f_{yx}f_{xy} > 0$. In this case, the function achieves its minimum.

d^2Z is negative definite iff $|H_1| = f_{xx} < 0$ And $|H_2| = f_{xx}f_{yy} - f_{yx}f_{xy} > 0$. This indicates that the function is at its maximum.

Considering the function of n- choice variables

$$Z = f(x_1, x_2, x_3, \dots, x_n)$$

The first order condition for the extremum of the function is $f_1 = f_2 = f_3 = \dots = 0$ which leads to the fact that $dZ = f_1dx_1 + f_2dx_2 + f_3dx_3 + \dots + f_ndx_n = 0$

The second order sufficient conditions are identified using the Hessian determinant

$$|H| = \begin{vmatrix} f_{11} & f_{12} & \dots & f_{1n} \\ f_{21} & f_{22} & \dots & f_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{n1} & f_{n2} & \dots & f_{nn} \end{vmatrix}$$

The sufficient condition for the maximum of the function is satisfied when $|H_1| < 0; |H_2| > 0; |H_3| < 0, \dots, (-1)^n |H_n| > 0$. where as for the minimum of the function all of the Hessian principal minors must be positive.

Example

1. Given the function $Z = 160x - 3x^2 - 2xy - 2y^2 + 120y - 18$, find the maximum value of the function.

Solution

The first order conditions that should be satisfied for maximum are

$$Z_x = \frac{\partial Z}{\partial x} = 0 \quad \text{And} \quad Z_y = \frac{\partial Z}{\partial y} = 0$$

$$Z_x = \frac{\partial Z}{\partial x} = 160 - 6x - 2y = 0$$

$$6x + 2y = 160 \quad \text{----- (1)}$$

$$Z_y = \frac{\partial Z}{\partial y} = -2x - 4y + 120 = 0$$

$$2x + 4y = 120 \quad \text{----- (2)}$$

Taking equation (1) and (2) simultaneously, multiplying (1) by 2 and subtracting equation (2) from this gives us

$$\begin{array}{r} \left\{ \begin{array}{l} 6x + 2y = 160 \\ 2x + 4y = 120 \end{array} \right. \quad \text{---} \\ \hline 12x + 4y = 320 \\ 2x + 4y = 120 \\ \hline 10x + 0 = 200 \\ 10x = 200 \\ x = 20 \end{array}$$

Substituting 20 in place of x in either of the two equation we will get $y = 20$

Taking the second partial derivatives,

$$Z_{xx} = -6 < 0, \quad Z_{yy} = -4 < 0$$

And

$$(Z_{xx})(Z_{yy}) > (Z_{xy})^2$$

$$(-6)(-4) > (-2)^2$$

$$24 > 4$$

$$\text{Alternatively, } |H| = \begin{vmatrix} -6 & -2 \\ -2 & -4 \end{vmatrix}, \quad |H_1| = -6 < 0 \quad \text{and} \quad |H_2| = 20 > 0$$

Thus, the function Z is maximized when $x = 20$ and $y = 20$

The maximum value of Z is

$$Z = 160(20) - 3(20)^2 - 2(20)(20) - 2(20)^2 + 120(20) - 18$$

$$Z = 2,782$$

3.2.1 Economic Applications

Example

1. A firm produces two products that are sold in two markets with the demand schedules $P_1 = 600 - 0.3Q_1$ and $P_2 = 500 - 0.2Q_2$. Production costs are related and the firm faces the total cost schedule $TC = 16 + 1.2Q_1 + 1.5Q_2 + 0.2Q_1Q_2$

- Determine the profit maximizing level of output and price in each market.
- Determine the maximum profit of the firm

Solution

$$\begin{aligned}
\text{Total Revenue (TR)} &= \text{TR}_1 + \text{TR}_2 \\
&= P_1 Q_1 + P_2 Q_2 \\
&= (600 - 0.3Q_1)Q_1 + (500 - 0.2Q_2)Q_2 \\
\text{TR} &= 600Q_1 - 0.3Q_1^2 + 500Q_2 - 0.2Q_2^2
\end{aligned}$$

$$\begin{aligned}
\text{Profit } (\pi) &= \text{TR} - \text{TC} \\
&= 600Q_1 - 0.3Q_1^2 + 500Q_2 - 0.2Q_2^2 - (16 + 1.2Q_1 + 1.5Q_2 + 0.2Q_1Q_2)
\end{aligned}$$

The first order conditions for maximum profit are

$$\begin{aligned}
\frac{\partial \pi}{\partial Q_1} &= 598.8 - 0.6Q_1 - 0.2Q_2 = 0 \\
0.6Q_1 + 0.2Q_2 &= 598.8 \text{----- (1)}
\end{aligned}$$

And

$$\begin{aligned}
\frac{\partial \pi}{\partial Q_2} &= 498.5 - 0.4Q_2 - 0.2Q_1 = 0 \\
0.2Q_1 + 0.4Q_2 &= 498.5 \text{----- (2)}
\end{aligned}$$

Taking equation (1) and (2) simultaneously, multiplying equation (2) by 3 deducting it from (1) gives us

$$\begin{aligned}
\text{Multiplying (2) by 3} \quad & \begin{cases} 0.6Q_1 + 0.2Q_2 = 598.8 \\ 0.2Q_1 + 0.4Q_2 = 498.5 \end{cases} \\
& 0.6Q_1 + 0.2Q_2 = 598.8 \\
& 0.6Q_1 + 1.2Q_2 = 1,495.5 \\
& -Q_2 = -896.7 \\
& Q_2 = 896.7 \text{ units}
\end{aligned}$$

Substituting this value for Q_2 in (1), we get

$$\begin{aligned}
598.8 - 0.6Q_1 - 0.2(896.7) &= 0 \\
598.8 - 179.34 &= 0.6Q_1 \\
419.46 &= 0.6Q_1 \\
Q_1 &= 699.1
\end{aligned}$$

To decide whether these output levels maximize the profit of the firm, We must check the second order conditions at these levels of output. That is

$$\left(\frac{\partial^2 \pi}{\partial Q_1^2} \right) = -0.6 < 0, \quad \frac{\partial^2 \pi}{\partial Q_2^2} = -0.4 < 0, \quad \frac{\partial^2 \pi}{\partial Q_1 \partial Q_2} = -0.2$$

$$\text{Therefore, } \left[\frac{\partial^2 \pi}{\partial Q_1^2} \right] \left[\frac{\partial^2 \pi}{\partial Q_2^2} \right] > \left[\frac{\partial^2 \pi}{\partial Q_1 \partial Q_2} \right]^2$$

$$\begin{aligned}
(-0.6)(-0.4) &> (-0.2)^2 \\
0.24 &> 0.04
\end{aligned}$$

The corresponding Hessian determinant is

$$|H| = \begin{vmatrix} -0.6 & -0.2 \\ -0.2 & -0.4 \end{vmatrix}, \quad |H_1| = -0.6 < 0; |H_2| = (-0.6)(-0.4) - (-0.2)^2 = 0.2 > 0$$

Therefore, profit is maximized when $Q_2 = 896.7$ and $Q_1 = 699.1$ and the profit maximizing prices are

$$\begin{aligned} P_1 &= 600 - 0.3(699.1) & P_2 &= 500 - 0.2(896.7) \\ &= 600 - 209.73 & &= 500 - 179.34 \\ P_1 &= 390.27 & P_2 &= 320.66 \end{aligned}$$

The maximum profit is

$$\begin{aligned} \pi &= 16 + 598.8(699.1) - 0.3(699.1)^2 + 498.5(896.7) - 0.2(896.7)^2 - 0.2(699.1) \\ &\quad (896.7) = 432,797.02 \text{ Birr} \end{aligned}$$

2. A multiplant monopoly operates two plants whose total cost functions are given by

$$TC_1 = 8.5 + 0.03 Q_1 \text{ and } TC_2 = 5.2 + 0.04 Q_2^2$$

If the demand function is given by $P = 60 - 0.04Q$, where $Q = Q_1 + Q_2$. How much output should the monopolist produce in each plant in order to maximize profit?

Solution

$$\text{Total Revenue (TR)} = P \times Q = (60 - 0.04Q)(Q)$$

$$TR = 60Q - 0.04Q^2$$

$$TR = 60(Q_1 + Q_2) - 0.04(Q_1 + Q_2)^2$$

$$TR = 60(Q_1 + Q_2) - 0.04(Q_1^2 + 2Q_1Q_2 + Q_2^2)$$

$$\text{Total Profit } (\pi) = TR - TC_1 - TC_2$$

$$\pi = -13.7 + 60Q_1 + 60Q_2 - 0.07Q_1^2 - 0.08Q_1Q_2 - 0.08Q_2^2$$

First order conditions for maximum profit are

$$\frac{\partial \pi}{\partial Q_1} = 60 - 0.14Q_1 - 0.08Q_2 = 0$$

$$0.14Q_1 + 0.08Q_2 = 60 \quad \text{----- (1)}$$

And

$$\frac{\partial \pi}{\partial Q_2} = 60 - 0.16Q_2 - 0.08Q_1 = 0$$

$$0.08Q_1 + 0.16Q_2 = 60 \quad \text{----- (2)}$$

Taking equation (1) and (2) simultaneously, multiplying (1) by 2 and reducing (2) from it

$$\begin{cases} 0.14Q_1 + 0.08Q_2 = 60 \\ 0.08Q_1 + 0.16Q_2 = 60 \end{cases}$$

$$0.28Q_1 + 0.16Q_2 = 120$$

$$\underline{0.08Q_1 + 0.16Q_2 = 60}$$

$$0.2Q_1 + 0 = 60$$

$$Q_1 = 300$$

Substituting this value in place of Q_1 in equation (1), we get

$$0.14(300) + 0.08Q_2 = 60$$

$$0.08Q_2 = 60 - 42$$

$$0.08Q_2 = 18$$

$$Q_2 = 225$$

When we check the second order conditions at these levels of output

$$\frac{\partial^2 \pi}{\partial Q_1^2} = -0.14 < 0, \quad \frac{\partial^2 \pi}{\partial Q_2^2} = -0.16 < 0, \quad \frac{\partial^2 \pi}{\partial Q_1 \partial Q_2} = -0.08$$

$$\text{Thus, } (-0.14)(-0.16) > (-0.08)^2$$

$$0.0224 > 0.0064$$

This verifies that profit is maximized when $Q_1 = 300$ and $Q_2 = 225$

The total amount of output that the firm should produce to maximize profit is

$$Q = Q_1 + Q_2 = 525 \text{ units.}$$

$$\text{Therefore, } P = 60 - 0.04(525) = 60 - 21 = 39$$

3. Suppose a firm faces the production function $Q = 0.8 K^{0.4} L^{0.3}$. It sells its output at a fixed price of 450 Birr a unit and can buy K and L at 15 Birr per unit and 8 Birr per unit respectively. What input mix will maximize profit?

Solution

$$\text{Total revenue (TR)} = PQ = 450(0.8K^{0.4}L^{0.3})$$

$$TR = 360 K^{0.4} L^{0.3}$$

$$\text{Total cost (TC)} = LP_L + KP_K$$

$$TC = 8L + 15K$$

$$\text{Thus total profit } (\pi) = TR - TC$$

$$\pi = 360 K^{0.4} L^{0.3} - 8L - 15K$$

The first order conditions for maximum output are

$$\frac{\partial \pi}{\partial L} = 108K^{0.4}L^{-0.7} - 8 = 0$$

$$108K^{0.4}L^{-0.7} = 8$$

$$\frac{K^{0.4}}{L^{-0.7}} = \frac{8}{108} \text{----- (1)}$$

$$\frac{\partial \pi}{\partial K} = 144K^{-0.6}L^{0.3} - 15 = 0$$

$$144K^{-0.6}L^{0.3} = 15$$

$$\frac{L^{0.3}}{K^{0.6}} = \frac{15}{144} \text{----- (2)}$$

Thus, we must have,

$$\text{From (1) } L^{0.7} = \frac{108}{8} K^{0.4}$$

$$L = \left[\frac{(108)}{8} \right]^{\frac{10}{7}} K^{\frac{4}{7}} \text{----- (3)}$$

$$\text{From (2) } K^{0.6} = \frac{144}{15} L^{0.3} \text{----- (4)}$$

Substituting equation (3) in place of L in (4), we get

$$K^{0.6} = \frac{144}{15} [(13.5)^{\frac{10}{7}} K^{\frac{4}{7}}]^{\frac{3}{10}}$$

$$K^{0.6} = \frac{144}{15} (13.5)^{\frac{3}{7}} K^{\frac{6}{35}}$$

$$K^{(0.6 - \frac{6}{35})} = \frac{144}{15} (13.5)^{\frac{3}{7}}$$

$$K^{\frac{3}{7}} = (9.6)(13.5)^{\frac{3}{7}}$$

$$K = (9.6)^{\frac{7}{3}} (13.5)$$

$$= (195.87) (13.5)$$

$$K = 2,644.25$$

Substituting the value of K in equation (3) enables us to determine the amount of labor that should be employed in the production process.

$$L = (13.5)^{\frac{10}{7}} (2,644.25)^{\frac{4}{7}}$$

$$= (41.19) (90.282)$$

$$L = 3,718.72$$

As you know it is necessary to check the second order conditions at these values. Therefore,

$$\frac{\partial^2 \pi}{\partial L^2} = -75.6 K^{0.4} L^{-1.7} = -75.6 (2,644.25)^{0.4} (3,718.72)^{-1.7}$$

$$= -0.0015 < 0$$

$$\frac{\partial^2 \pi}{\partial K^2} = -86.4 K^{-1.6} L^{0.3} = -86.4 (2,644.25)^{-0.6} (3,718.72)^{-0.7}$$

$$= 0.00121$$

$$\left(\frac{\partial^2 \pi}{\partial L^2}\right)\left(\frac{\partial^2 \pi}{\partial K^2}\right) > \left(\frac{\partial^2 \pi}{\partial L \partial K}\right)^2$$

$$\text{Therefore, } (-0.0015)(-0.0034) > (0.00121)^2$$

The corresponding Hessian determinant is

$$|H| = \begin{vmatrix} -0.0015 & 0.00121 \\ 0.00121 & -0.0034 \end{vmatrix}, \quad |H_1| = -0.0015 < 0; |H_2| > 0.$$

Now we are confident enough to conclude that the firm will maximize its profit when it employs 2,644.25 units of capital and 3,718.72 units of labor in the production process.

4. suppose the monopolist sells a certain product in three separate markets and the demand functions facing the firm are

$$P_1 = 63 - Q_1$$

$$P_2 = 105 - 5Q_2 \quad \text{And the cost function is } C = 20 - 15Q; Q = Q_1 + Q_2 + Q_3$$

$$P_3 = 75 - 6Q_3$$

Determine the amount of output that should be sold to maximize profit. Identify the prices charged in each market to maximize profit.

First we should construct the total revenues for each market. These are

$$TR_1 = 63Q_1 - Q_1^2$$

$$TR_2 = 105Q_2 - 5Q_2^2$$

$$TR_3 = 75Q_3 - 6Q_3^2$$

Therefore, the total profit function is $\pi = TR_1 + TR_2 + TR_3 - C$

$$\pi = 63Q_1 - Q_1^2 + 15Q_2 - 5Q_2^2 + 75Q_3 - 6Q_3^2 - (20 + 15Q_1 + 15Q_2 + 15Q_3)$$

$$\pi = 48Q_1 - 4Q_1^2 + 90Q_2 - 5Q_2^2 + 60Q_3 - 6Q_3^2 - 20 \text{-----(1)}$$

First order conditions

$$\pi_1 = \frac{\partial \pi}{\partial Q_1} = 48 - 8Q_1 = 0 \Rightarrow Q_1 = 6$$

$$\pi_2 = 90 - 10Q_2 = 0 \Rightarrow Q_2 = 9$$

$$\pi_3 = 60 - 12Q_3 = 0 \Rightarrow Q_3 = 5$$

$$P_1 = 57, P_2 = 60, P_3 = 45$$

We can check the second order condition using the sign of the principal minors of the Hessian determinant. The corresponding Hessian determinate is

$$\pi_{11} = -8, \pi_{12} = 0, \pi_{13} = 0$$

$$\pi_{21} = 0, \pi_{22} = -10, \pi_{23} = 0$$

$$\pi_{31} = 0, \pi_{32} = 0, \pi_{33} = -12$$

$$|H| = \begin{vmatrix} \pi_{11} & \pi_{12} & \pi_{13} \\ \pi_{21} & \pi_{22} & \pi_{23} \\ \pi_{31} & \pi_{32} & \pi_{33} \end{vmatrix} = \begin{vmatrix} -8 & 0 & 0 \\ 0 & -10 & 0 \\ 0 & 0 & -12 \end{vmatrix}$$

Evaluating the successive principal minors,

$$|H_1| = -8 < 0, |H_2| = \begin{vmatrix} -8 & 0 \\ 0 & -10 \end{vmatrix} = 80 > 0, |H_3| = |H| = -960 < 0$$

It indicates that $d^2\pi$ is negative definite. Thus, the monopolist can maximize its profit when it sells 20 units of output of which 6 units at a price of 57 birr per unit in market 1, 9 units at a price of 60 birr per unit in market 2 and 5 units at a price of 45 birr per unit in market 3.

Dear colleague! By now you have completed the second section of this unit. Therefore, try to do the following self - test questions to evaluate whether you understand the section or not.

Section 3.3 Unconstrained Envelope Theorems

Until now we have discussed about the way how the maximum or minimum value of the function depends on the value of the independent variables under consideration. However, in economic theory we are usually interested in how the optimal value of a function depends on some parameters such as tax rates etc.

Although these parameters are assumed to be constant during the process of optimization, they may vary according to the economic situation. Therefore, what happens to the optimal value of the objective function when these parameters change?

Considering the firm which produces an output Q using L units of labor as input, and its production function is given by $Q = f(L)$. Suppose the price of the product is p and that of labor is w , the theory of the firm states that the firm chooses the amount of labor L which maximizes profit. The profit function is

$$\pi[L, (w, p)] = pf(L) - wL$$

Considering $L^*(w, p)$ as the optimal amount of labor when the prices are w and p , then the maximal profit of the firm is represented by

$$\pi^*(w, p) = pf[L^*(w, p)] - wL^*(w, p)$$

This function is known as the firm's **indirect profit function**. According to the envelope theory, the partial derivative of the profit function with respect to p , evaluated at $L = L^*(w, p)$ is $\pi(L^*(w, p))$

In this case the derivative is positive which indicates that as the **price of the product** increases, the profit of the firm increases.

Similarly, the envelope theorem states that the derivative of the firm's profit function with respect to w is $-L^*(w, p)$. But in this case the derivative is negative which shows that an increase in the **price of an input decreases** the maximal profit of the firm.

Dean Colleague! The following graph shows how the indirect objective function envelopes the direct objective function.

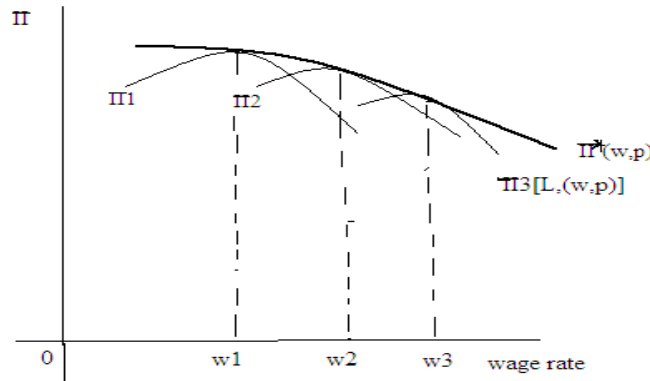


Fig.3.1 Envelop Curve.

Each of the inverted U-shaped curves is the graph of direct profit function $\pi(L, (w, p))$ as a function of w for a given value of L and p . Each of these graphs shows how π changes due to change in w for a given value of L and p . To get the solution of the maximization problem for any given value of w , we should find the highest function for that value of w . Thus the graph of the indirect profit function is the locus of these highest points. Thus, the indirect objective function is the envelop curve of the direct objective function at various values of w in our case.

Example

1. Suppose a firm is producing a certain product Q and wants to maximize its profit. Suppose a tax rate t is imposed on a production of Q . What is the effect of change in the tax rate on total profit?

$$\begin{aligned} \text{Total Profit } (\pi) &= TR - TC \\ TR &= R(Q) \\ TC &= C(Q) + tQ \end{aligned}$$

$$\text{Thus, } \pi(Q, t) = R(Q) - C(Q) - tQ$$

The first order condition for maximum profit is

$$\pi'(Q, t) = R'(Q) - C'(Q) - t = 0$$

$$MR - MC - t = 0$$

$$MR = MC + t \text{----- (1)}$$

From equation (1) we can determine the critical value of the profit function and let's say $Q = Q^*$.

Second order condition for maximum profit is

$$\pi''(Q, t) = R''(Q) - C''(Q) < 0$$

Given the optimal output $Q = Q^*(t)$, the maximal profit is

$$\pi(Q^*(t)) = R(Q^*(t)) - C(Q^*(t)) - tQ^*(t)$$

The effect of change in the tax rate on total profit is determined by differentiating the

Profit function with respect to the tax rate. It is given as

$$\begin{aligned} \frac{d\pi}{dt} &= R'(Q^*(t)) \cdot \frac{dQ^*}{dt} - C'(Q^*(t)) \cdot \frac{dQ^*}{dt} - Q^* - t \frac{dQ^*}{dt} \\ &= [R'(Q^*(t)) - C'(Q^*(t))] \frac{dQ^*}{dt} - Q^* - t \frac{dQ^*}{dt} \end{aligned}$$

$$\text{However, } R'(Q^*(t)) = C'(Q^*(t)) + t$$

$$\text{Therefore, } \frac{d\pi}{dt} = [C'(Q^*(t)) + t - C'(Q^*(t))] \frac{dQ^*}{dt} - Q^* - t \frac{dQ^*}{dt}$$

$$= t \frac{dQ^*}{dt} - Q^* - t \frac{dQ^*}{dt}$$

$$\frac{d\pi}{dt} = -Q^*$$

As you have seen the rate of change of total profit with respect to tax rate is negative. It indicates that an increase in tax decreases the total profit of the firm.

1. Assume that the demand and the total cost functions of the monopolist are $P=24-3x$ and $C=x^2+8x$ respectively. Determine the rate of change of the profit function with respect to the tax rate when a tax rate of 4 Birr per unit of production is imposed.

$$\text{Total profit } \pi = (24 - 3x)x - (x^2 + 8x + 4x)$$

$$= 24x - 3x^2 - x^2 - 12x$$

$$\pi = -4x^2 + 12x$$

Now let us determine the critical value applying the first order condition.

$$\pi' = -8x + 12 =$$

$$-8x = -12$$

$$x^* = 1.5 \text{ units}$$

Second order condition is

$$\pi'' = -8 < 0$$

Therefore, $x = x^* = 1.5$ is the optimal amount of output.

$$\text{Then } \pi(t) = -4x^2 + 16x - 1.5t = -4(1.5)^2 + 16(1.5) - 1.5t$$

$$= -9 + 24 - 1.5t = 15 - 1.5t$$

$$\frac{d\pi}{dt} = -1.5 = -x^*$$